

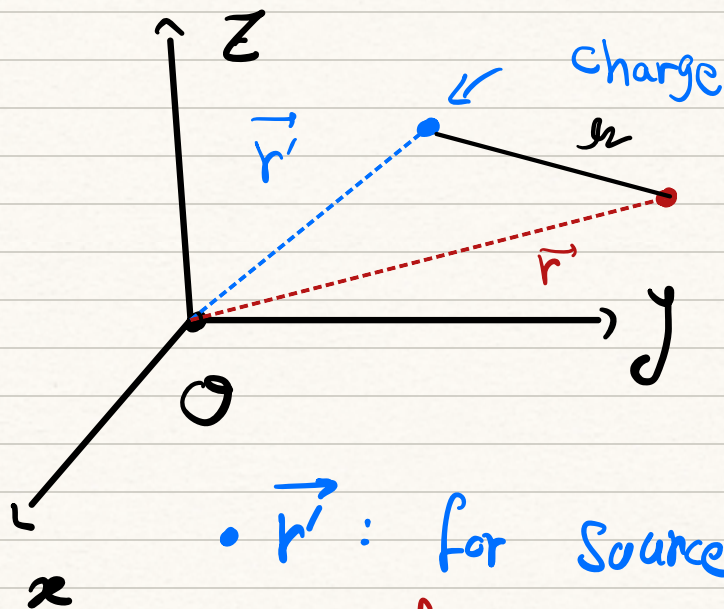
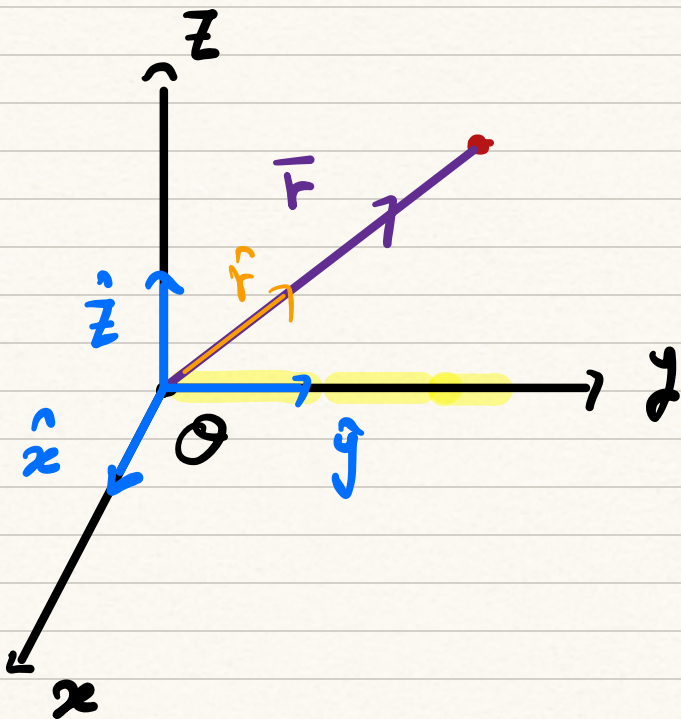
Chap 1: Vector Calculus

• Unit Vectors $|\hat{x}| = |\hat{y}| = |\hat{z}| = 1$

$$\vec{r} = x \hat{x} + y \hat{y} + z \hat{z}$$

$$r = |\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

$$\hat{r} = \frac{\vec{r}}{r}$$



effect

• $\vec{r}'$: for Source Point

• $\vec{r}$: for Point of InterSet.

$\vec{r}$: Separation vector.

$$\vec{r} \equiv \vec{r} - \vec{r}'$$

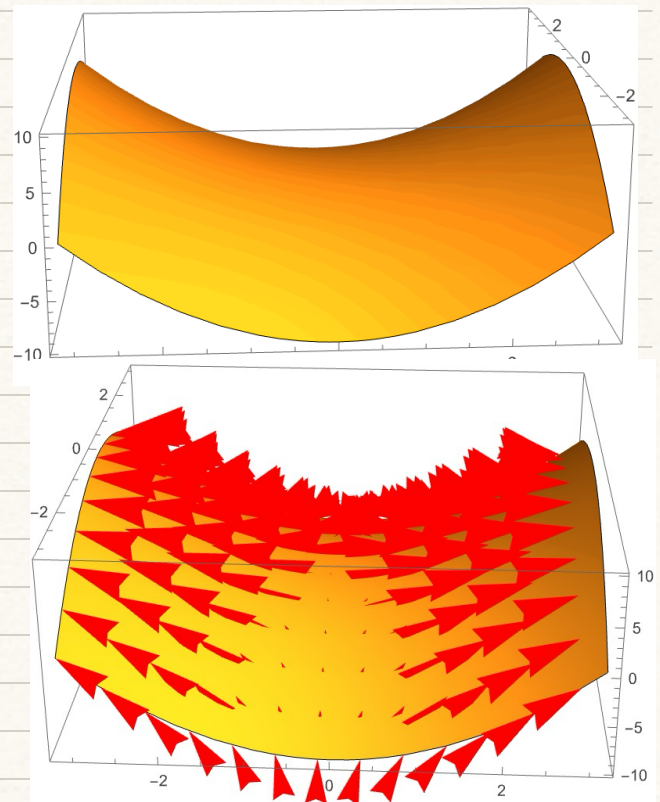
Sec1 | Differential Operators

1) Gradient: ∇T

Apply on a scalar field to produce a vector
Points to the largest change in T

$$\nabla T = \left(\frac{\partial T}{\partial x}, \frac{\partial T}{\partial y}, \frac{\partial T}{\partial z} \right)$$

Exp 1.1) let $T = x^2 - y^2$, find ∇T

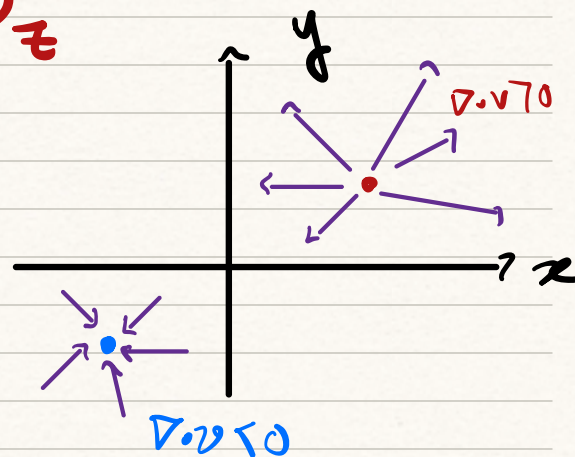


2) The Divergence: $\nabla \cdot \vec{v}$

Apply on a vector field to Produce a Scalar

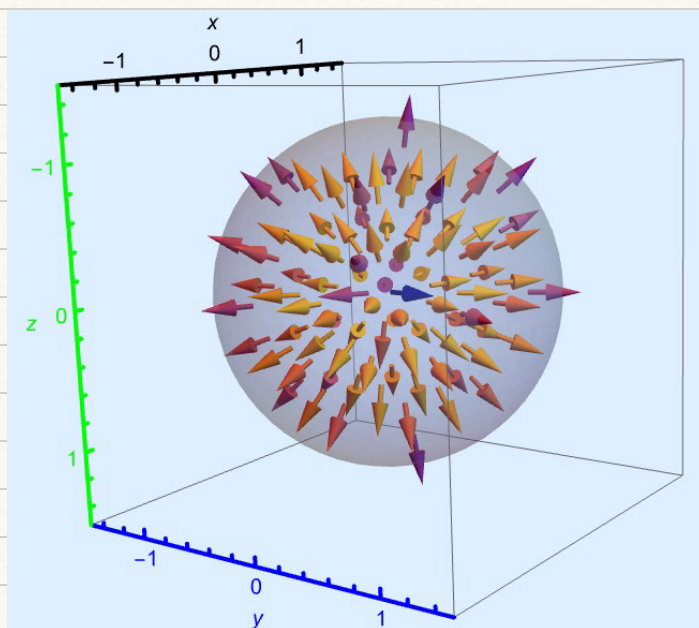
measures the spreading of $\vec{v}$.

$$\nabla \cdot \vec{v} = \frac{\partial}{\partial x} v_x + \frac{\partial}{\partial y} v_y + \frac{\partial}{\partial z} v_z$$



Exp 1.2) : $\vec{v} = e^{-(x^2+y^2+z^2)} (x \hat{x} + y \hat{y} + z \hat{z})$

Find $\nabla \cdot \vec{v}$

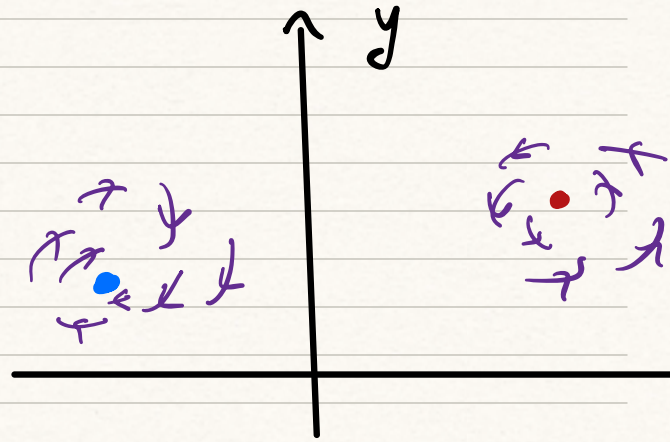


3) The Curl: $\nabla \times \vec{v}$

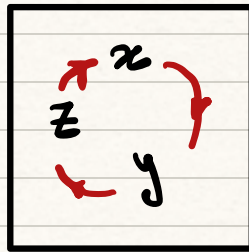
Apply on a vector field to Produce a Vector

Points to the axis of rotation of $\vec{v}$

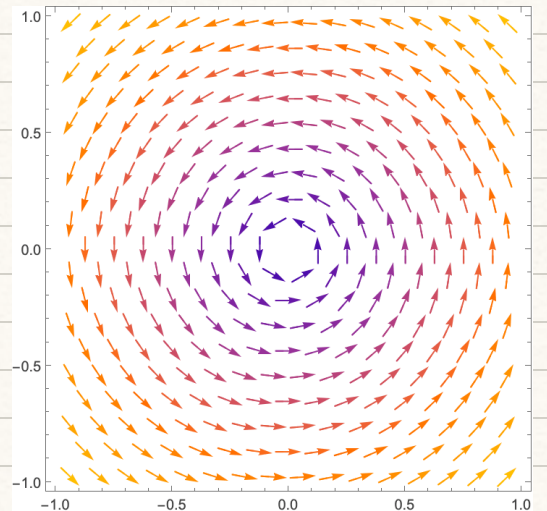
$$\nabla \times \vec{v} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ v_x & v_y & v_z \end{vmatrix}$$



$$= (\partial_y v_z - \partial_z v_y) \hat{x} + (\partial_z v_x - \partial_x v_z) \hat{y} + (\partial_x v_y - \partial_y v_x) \hat{z}$$



EXP 1.3: $\vec{v} = -2y \hat{x} + 2x \hat{y}$



Sec2

Fundamental Theorems

Bank account:

Initial $\text{£} 500$

$+\text{£} 50$

$-\text{£} 10$

$-\text{£} 20$

$+\text{£} 60$

$-\text{£} 30$

Final: $\text{£} 550$

Q: How much did we gain?

$$\begin{aligned} A1) & +50 - 10 - 20 + 60 - 30 \\ & = 50 \end{aligned}$$

$$A2) 550 - 500 = 50$$

$\Rightarrow$ The lesson is:

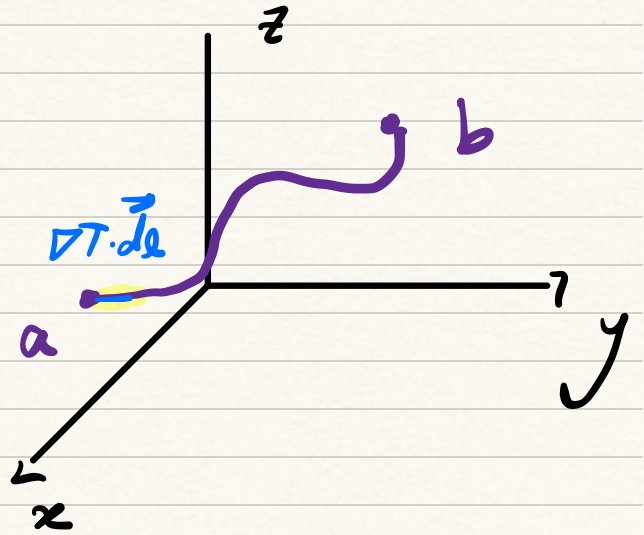
The sum of the small changes inside

$=$

The big change at boundary

1) Gradients:

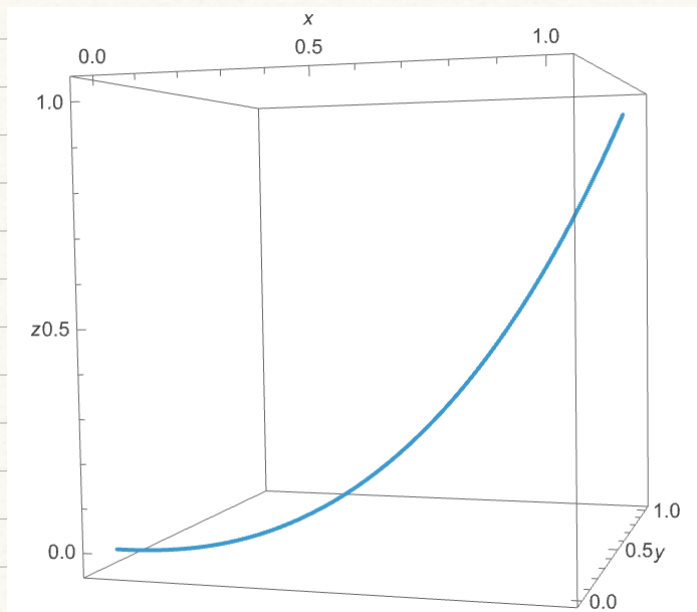
$$\int \nabla T \cdot d\vec{l} = T(b) - T(a)$$



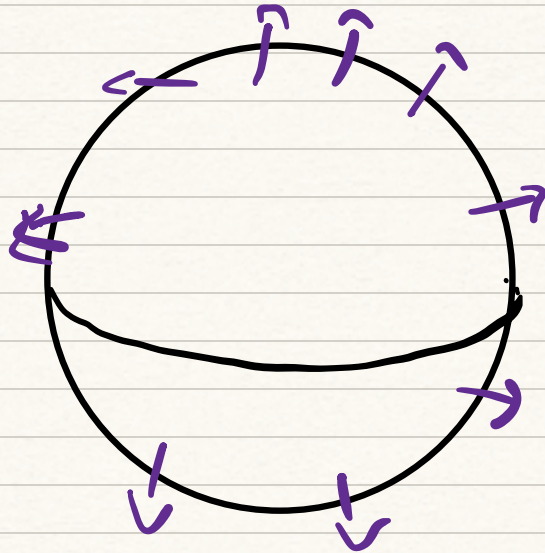
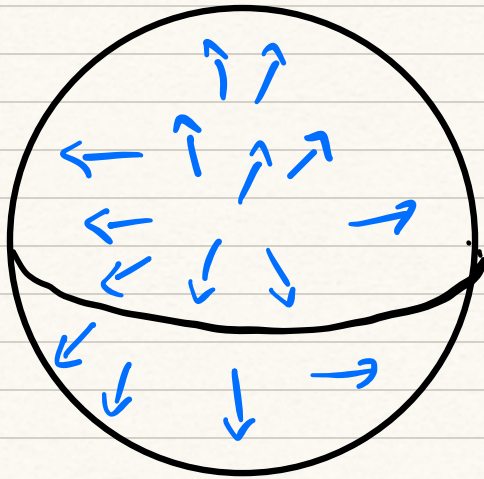
Exp 2.1) $T = x^2y + yz^2$, $a = (0,0,0)$, $b = (1,1,1)$

along $y = x^2$, $z = x$; $0 \leq x \leq 1$

Check $\int \nabla T \cdot d\vec{l} = T(b) - T(a)$



2) Divergences



$$\iiint_V \nabla \cdot \vec{v} \, d\tau = \oiint_S \vec{v} \cdot d\vec{a}$$

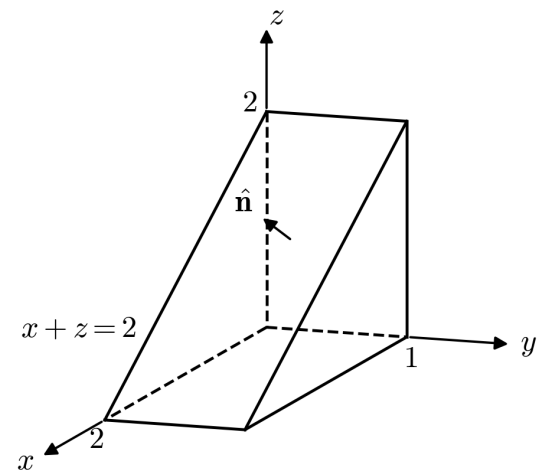
Sum of small divergences inside

=

One big divergences at the boundary, i.e.
Flux.

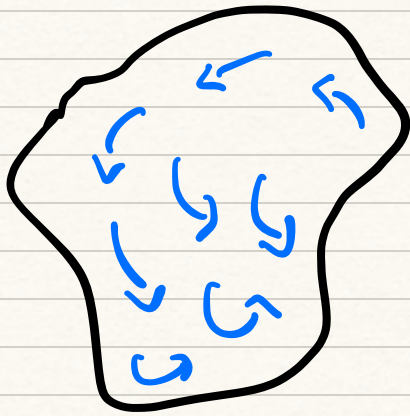
EXP2.2) $\vec{v} = x \hat{x} + 2y \hat{y} + z \hat{z}$

check: $\iiint \nabla \cdot \vec{v} = \oiint \vec{v} \cdot d\vec{a}$



$$0 \leq y \leq 1, \quad x \geq 0, \quad z \geq 0, \quad x + z \leq 2$$

3) Stoke's Theorem



$$\iint_A (\nabla \times \vec{v}) \cdot d\vec{a} = \oint_P \vec{v} \cdot d\vec{l}$$

Sum of small circulations inside

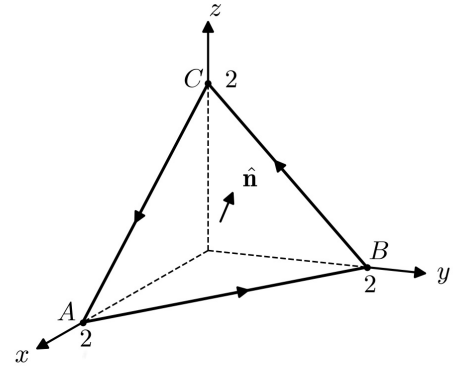
=

one big circulation at the boundary

Gradients	$\int \nabla T \cdot d\vec{l}$	$T(b) - T(a)$	line integral $\Rightarrow$ diff end points
Div	$\iiint \nabla \cdot \vec{v}$	$\oiint \vec{v} \cdot d\vec{a}$	Volume integral = closed surface
Stoke's	$\iint (\nabla \times \vec{v}) \cdot d\vec{a}$	$\oint \vec{v} \cdot d\vec{l}$	Surface integral = closed path

Exp 2.3) $\vec{v} = z \hat{x} + x \hat{y} + y \hat{z}$

check $\iint (\nabla \times \vec{v}) \cdot d\vec{a} = \oint \vec{v} \cdot d\vec{l}$



$$x + y + z = 2, \quad x, y, z \geq 0$$

Sec 3 | Coordinates Systems

1) Cartesian Coordinates:

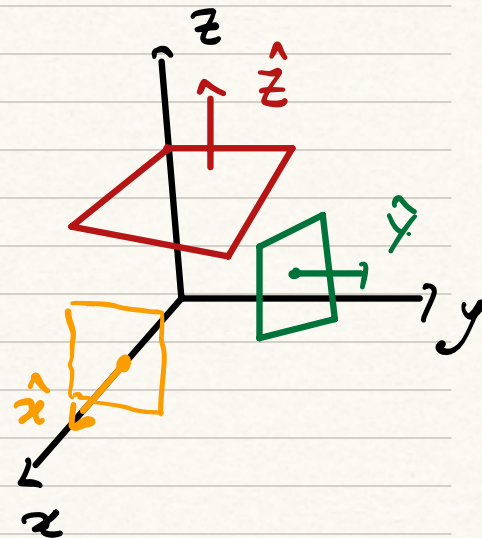
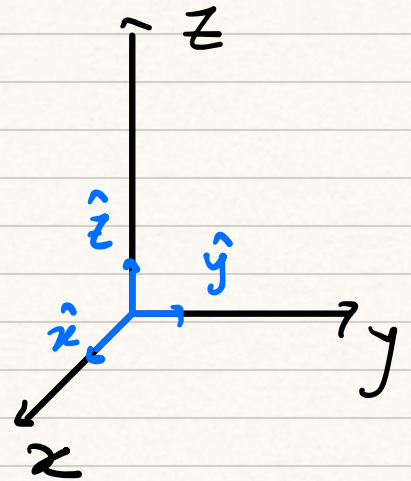
$$d\vec{l} = dx \hat{x} + dy \hat{y} + dz \hat{z}$$

$$d\vec{a} = dx dy \hat{z} \quad \text{Fix } z$$

$$d\vec{a} = dx dz \hat{y} \quad \text{Fix } y$$

$$d\vec{a} = dy dz \hat{x} \quad \text{Fix } x$$

$$dV = dx dy dz$$

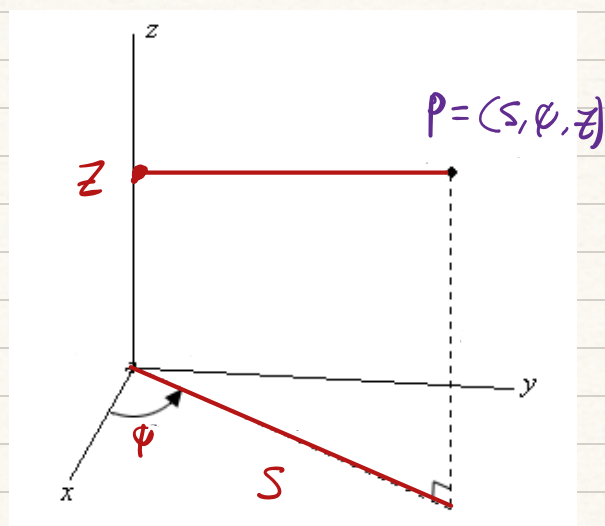


2) Cylindrical Coordinates:

$$x = s \cos \phi, \quad 0 \leq \phi < 2\pi$$

$$y = s \sin \phi$$

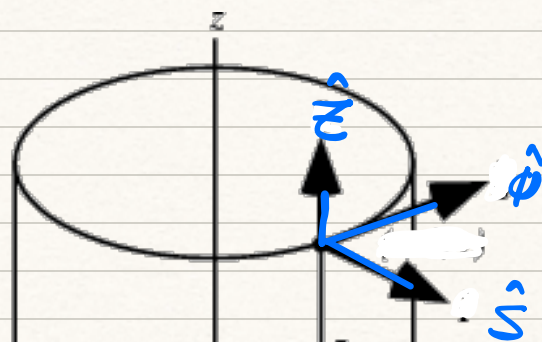
$$z = z$$



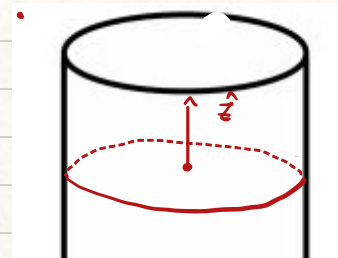
$$d\vec{r} = ds \hat{s} + s d\phi \hat{\phi} + dz \hat{z}$$

* why $s d\phi$?

$$s d\phi = s d\phi$$



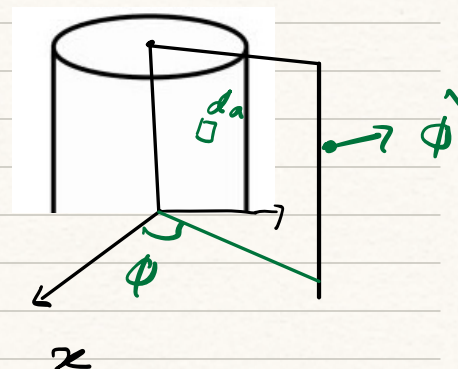
$$d\vec{a} = s d\phi ds \hat{z} \quad \text{Fix } z$$



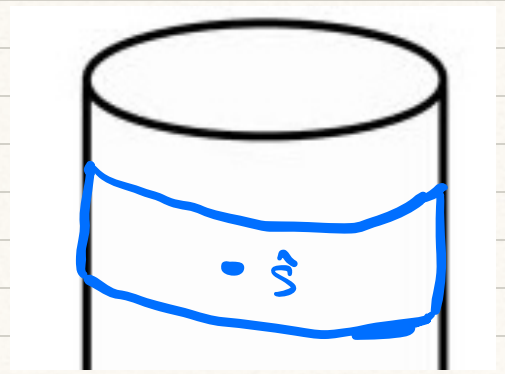
$$d\vec{a} = ds dz \hat{\phi} \quad \text{Fix } \phi$$

$$\hat{\phi} = -\sin \phi \hat{x} + \cos \phi \hat{y}$$

$\phi = 0$	$\phi = \frac{\pi}{2}$	$\phi = \pi$
$\hat{\phi} = \hat{y}$	$\hat{\phi} = -\hat{x}$	$\hat{\phi} = -\hat{y}$



$$d\vec{a} = s d\phi dz \vec{s} \quad \text{Fix } s$$



$$\Rightarrow d\tau = s ds d\phi dz$$

3) Spherical Coordinates:

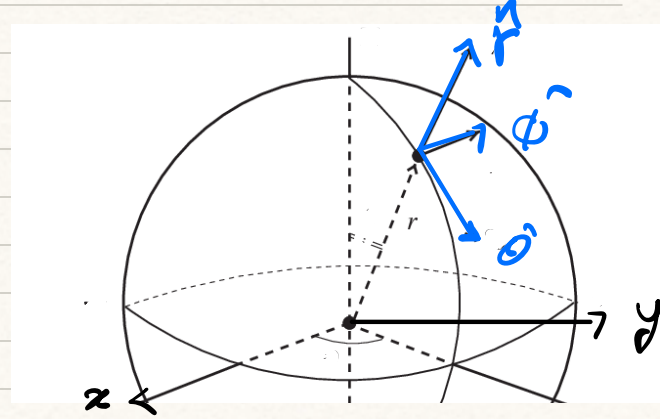
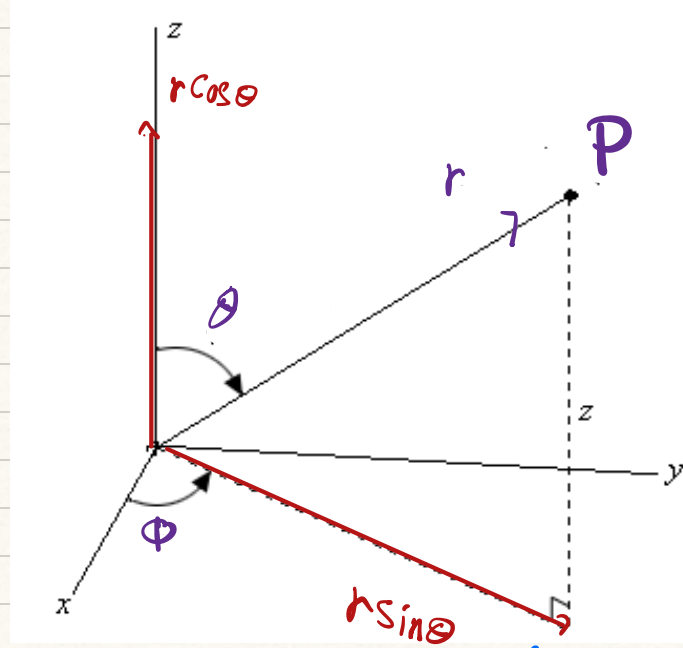
$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

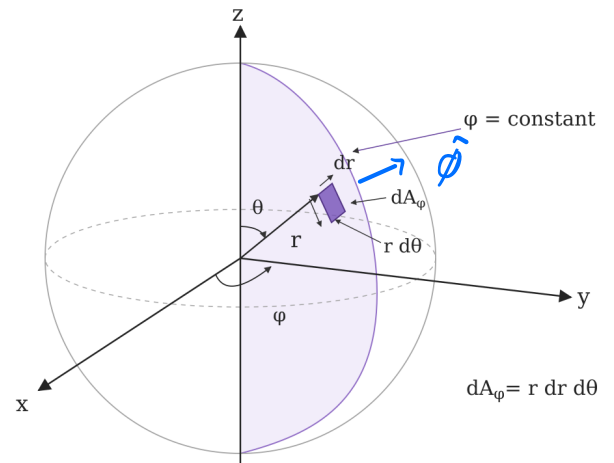
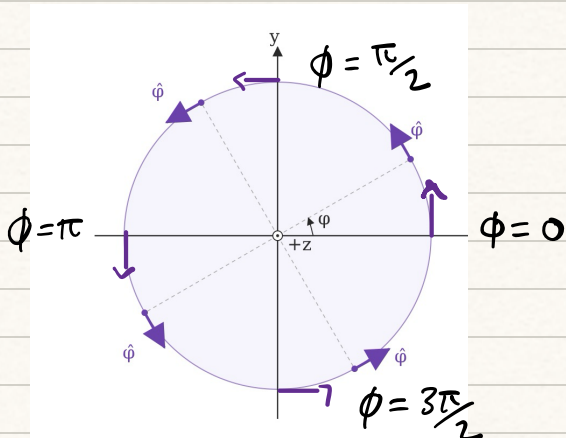
$$0 \leq \phi < 2\pi, \quad 0 \leq \theta \leq \pi$$

$$d\vec{r} = dr \hat{r} + r d\theta \hat{\theta} + r \sin \theta d\phi \hat{\phi}$$

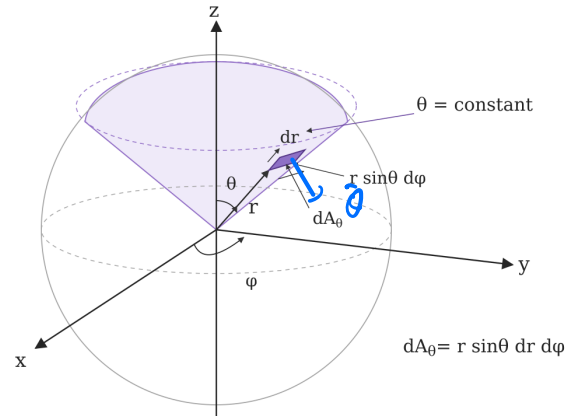


$$d\vec{a} = r dr d\theta \hat{\phi} \quad \text{Fix } \phi$$

$$\hat{\phi} = -\sin \phi \hat{x} + \cos \phi \hat{y}$$



$$d\vec{a} = r \sin \theta \, dr \, d\phi \quad \hat{\theta} \quad \text{Fix } \theta$$



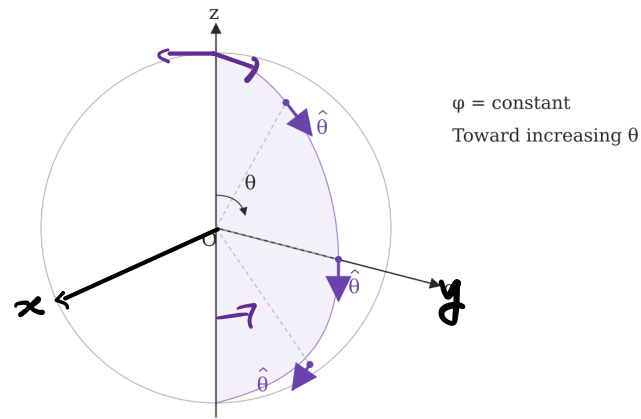
$$\hat{\theta} = \cos \theta \cos \phi \hat{x} + \cos \theta \sin \phi \hat{y} - \sin \theta \hat{z}$$

Case: $\theta = 0$

$$\phi = 0 \quad \hat{\theta} = \hat{z}$$

$$\phi = \frac{\pi}{2} \quad \hat{\theta} = \hat{y}$$

Case: $\theta = \frac{\pi}{2}$, $\hat{\theta} = -\hat{z}$



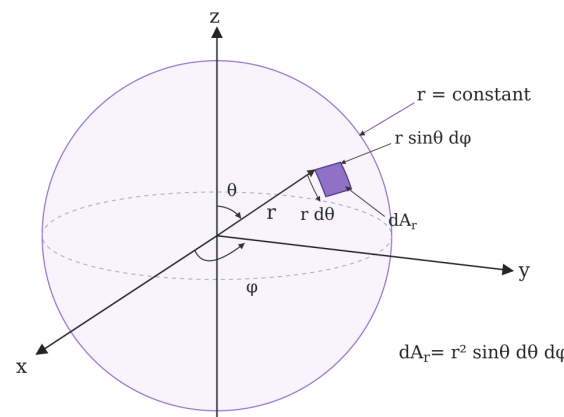
Case: $\theta = \pi$

$$\phi = 0 \quad \hat{\theta} = -\hat{z}$$

$$\phi = \frac{\pi}{2} \quad \hat{\theta} = -\hat{y}$$

$$d\vec{a} = r^2 \sin \theta \, d\theta \, d\phi \quad \hat{r} \quad \text{Fix } r$$

$$d\tau = r^2 \sin \theta \, dr \, d\theta \, d\phi$$



Sec 4 | Fundamental Theorems In

Curvilinear Coordinates:

• Cylindrical Coordinates:

$$\vec{\nabla} T = \left(\frac{\partial T}{\partial s} \right) \hat{s} + \frac{1}{s} \left(\frac{\partial T}{\partial \phi} \right) \hat{\phi} + \left(\frac{\partial T}{\partial z} \right) \hat{z}$$

$$\vec{\nabla} \cdot \vec{v} = \frac{1}{s} \frac{\partial}{\partial s} (s v_s) + \frac{1}{s} \frac{\partial}{\partial \phi} (v_\phi) + \frac{\partial v_z}{\partial z}$$

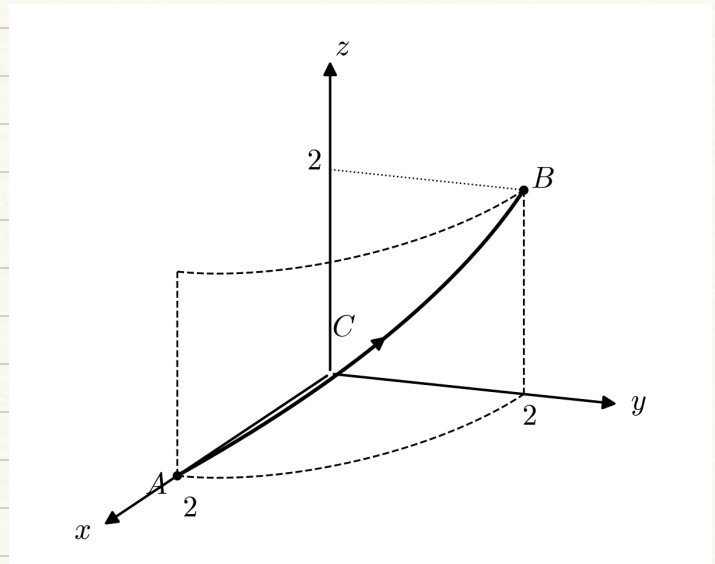
$$\vec{\nabla} \times \vec{v} = \left(\frac{1}{s} \frac{\partial v_z}{\partial \phi} - \frac{\partial v_\phi}{\partial z} \right) \hat{s} + \left(\frac{\partial v_s}{\partial z} - \frac{\partial v_z}{\partial s} \right) \hat{\phi} + \frac{1}{s} \left(\frac{\partial}{\partial s} (s v_\phi) - \frac{\partial v_s}{\partial \phi} \right) \hat{z}$$

Ex 4.1) $T = s^2 \sin \phi + z^2$. $A = (2, 0, 0) \rightarrow B = (2, \pi/2, 2)$

Along the path

$$s = 2, \quad z = \frac{4}{\pi} \phi, \quad 0 \leq \phi \leq \frac{\pi}{2}$$

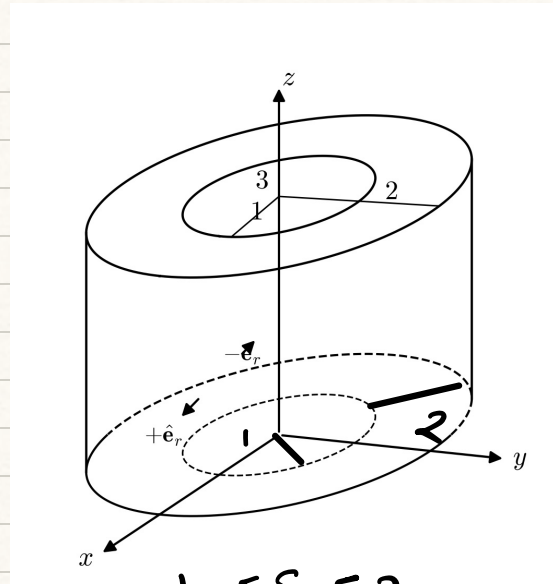
check $\int \nabla T \cdot d\vec{r} = T(B) - T(A)$



$$\text{Exp 4.2) } \vec{v} = s z \hat{s} + s^2 \hat{\phi} + z^2 \hat{z}$$

check that

$$\iiint \nabla \cdot \vec{v} d\tau = \oint \vec{v} \cdot d\vec{a}$$

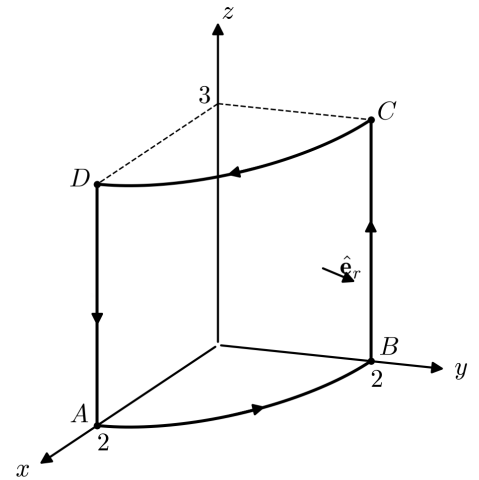


$$1 \leq s \leq 2$$

$$0 \leq z \leq 3$$

Ex 4.3) $\vec{v} = 5(z+1) \hat{\phi} + 5^2 \hat{z}$

check that $\iint_P \nabla \times \vec{v} \cdot d\vec{a} = \oint_P \vec{v} \cdot d\vec{x}$



• Spherical Coordinates:

$$\vec{\nabla} T = \frac{\partial T}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial T}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial T}{\partial \phi} \hat{\phi}$$

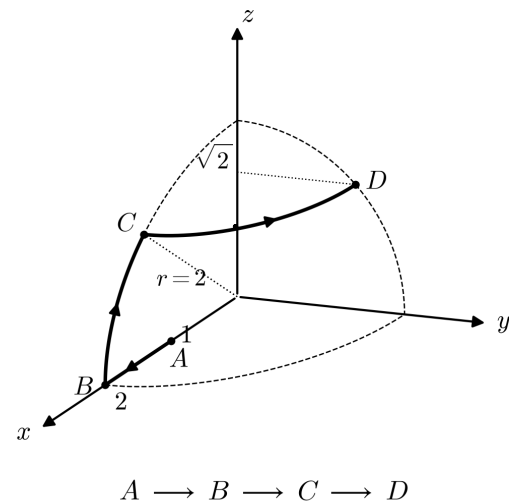
$$\vec{\nabla} \cdot \vec{v} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 v_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta v_\theta) + \frac{1}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi}$$

$$\vec{\nabla} \times \vec{v} = \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (\sin \theta v_\phi) - \frac{\partial v_\theta}{\partial \phi} \right] \hat{r} + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial v_r}{\partial \phi} - \frac{\partial}{\partial r} (r v_\phi) \right] \hat{\theta} + \frac{1}{r} \left[\frac{\partial}{\partial r} (r v_\theta) - \frac{\partial v_r}{\partial \theta} \right] \hat{\phi}$$

Exp 4.4) $T = r^2 \cos \theta + r \sin \theta \cos \phi$

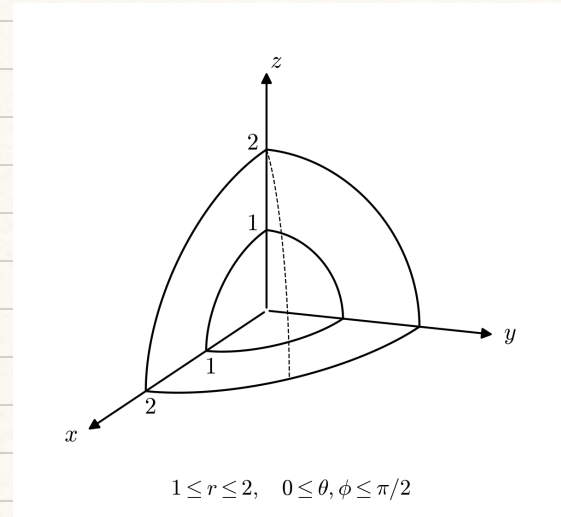
check $\int \nabla T \cdot d\vec{l} = T(D) - T(A)$

$A = (1, \frac{\pi}{2}, 0), D = (2, \frac{\pi}{4}, \frac{\pi}{2})$



Exp 4.5) $\vec{u} = r^2 \hat{r} + r \sin \theta \hat{\phi}$

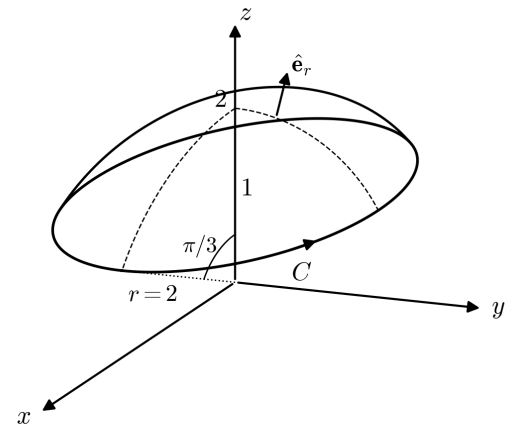
check $\iiint \nabla \cdot \vec{u} \, d\tau = \oiint \vec{u} \cdot d\vec{a}$



ExP 4.6)

$$\vec{v} = r \sin \theta \hat{\phi}$$

Show $\iint (\nabla \times \vec{v}) \cdot d\vec{a} = \oint \vec{v} \cdot d\vec{r}$



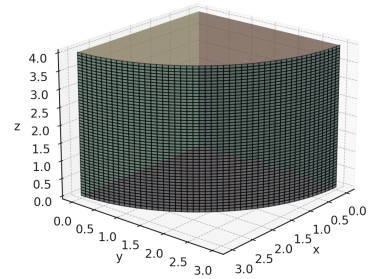
$$r=2, \quad 0 \leq \theta \leq \pi/3$$

Extra Problems:

P1) $\vec{v} = s^2 \cos \phi \hat{s} + s \sin \phi \hat{\phi} + z^2 \hat{z}$

Show $\iiint_V \nabla \cdot \vec{v} \, d\tau = \oiint_S \vec{v} \cdot d\vec{a}$

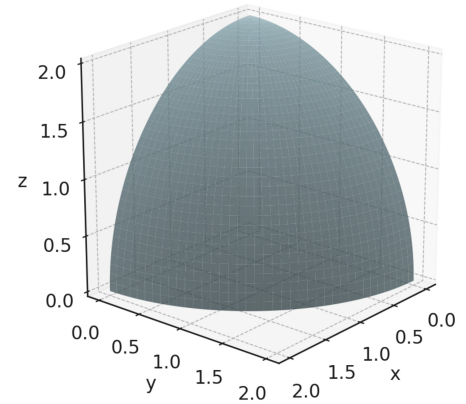
Quarter-cylinder region: $0 \leq s \leq 3$, $0 \leq \phi \leq \pi/2$, $0 \leq z \leq 4$



P2) $\vec{V} = r^2 \sin\theta \hat{r} + r \cos\theta \hat{\theta} + r \sin\theta \sin\phi \hat{\phi}$

Show $\iiint \nabla \cdot \vec{V} d\tau = \oiint \vec{V} \cdot d\vec{a}$

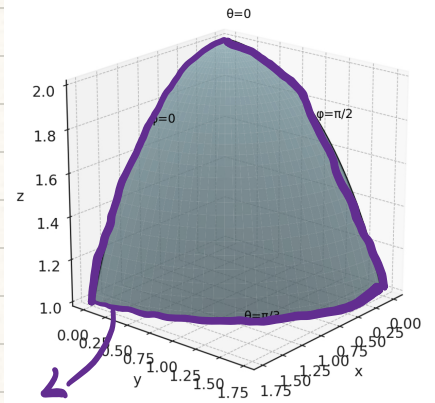
Spherical Octant ($0 \leq r \leq R$, $0 \leq \theta \leq \pi/2$, $0 \leq \phi \leq \pi/2$)



P3) $\vec{V} = \frac{1}{2} r \sin \theta \hat{\phi}$

Show $\iint_P (\nabla \times \vec{v}) \cdot d\vec{a} = \oint_P \vec{v} \cdot d\vec{x}$

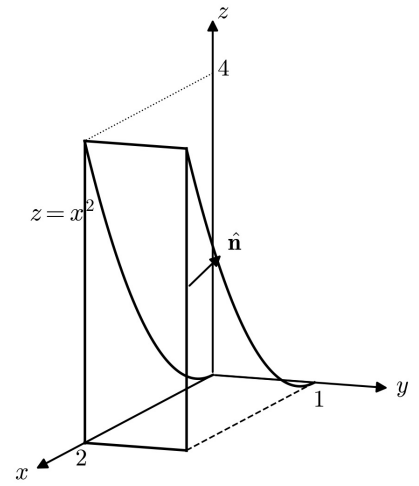
Spherical rectangle: $R=2$, $0 \leq \theta \leq \pi/3$, $0 \leq \phi \leq \pi/2$



P

P4) $\vec{V} = x \hat{x} + y \hat{y} + z \hat{z}$

Find $\oiint \vec{V} \cdot d\vec{a}$



$$0 \leq x \leq 2, \quad 0 \leq y \leq 1, \quad 0 \leq z \leq x^2$$